

Centroids by Integration

① Basic Definitions:

$$\bar{x} = \frac{\int_A \tilde{x} dA}{\int_A dA}$$

$$\bar{y} = \frac{\int_A \tilde{y} dA}{\int_A dA}$$

② Choices for dA :

$$dA = dx dy$$

$$dA = dy dx$$

Horiz
Element

$$dA = \left[\begin{array}{c} \text{---} \frac{dx}{dy} dy \\ x_L \text{ as } f(y) \quad x_R \text{ as } f(y) \end{array} \right]$$

Since dy ,
Define
 x_L and x_R
in terms of y

$$dA =$$

Vert
Element

$$\left[\begin{array}{c} y_{\text{top}} \text{ as } f(x) \\ \text{---} dx \\ y_{\text{bot}} \text{ as } f(x) \end{array} \right]$$

Since dx ,
Define y_{top} + y_{bot}
in terms of x

③ Double Integrals, limits must corresp to choice of dA :

$$\int_{x_L^* \text{ as } f(y)}^{x_R^* \text{ as } f(y)} \int_{y_{\text{bot}}^* \text{ as } f(x)}^{y_{\text{top}}^* \text{ as } f(x)} dx dy \quad * \text{ in terms of } y \text{ (as } f(y))$$

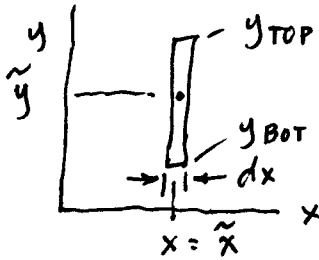
$$\text{or} \quad \int_{y_{\text{bot}}^+ \text{ as } f(x)}^{y_{\text{top}}^+ \text{ as } f(x)} \int_{x_L^+ \text{ as } f(y)}^{x_R^+ \text{ as } f(y)} dy dx \quad + \text{ in terms of } x \text{ (as } f(x))$$

④ Choices for $\tilde{x}$, $\tilde{y}$: These are the locations of the centroids of the chosen dA 's.

(a) Double Integral (Easy!): $\tilde{x} = x$ Regardless of
 $\tilde{y} = y$ $dA = dx dy$
or $dA = dy dx$

④ (cont'd) Choice of $\tilde{x}$ or $\tilde{y}$

(b) Vertical Element



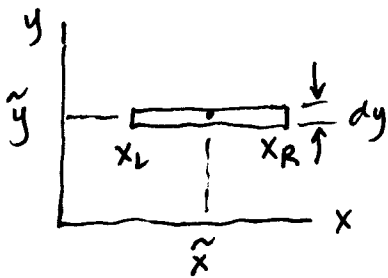
$$\tilde{x} = x$$

$$\tilde{y} = \frac{1}{2}(y_{TOP}^* + y_{BOT}^*)$$

$$dA = (y_{TOP}^* - y_{BOT}^*) dx$$

* Defined
in terms
of x

(c) Horiz Element



$$\tilde{y} = y$$

$$\tilde{x} = \frac{1}{2}(x_L^* + x_R^*)$$

$$dA = (x_R^* - x_L^*) dy$$

+ Defined
in terms
of y